The problem of meaning in mathematics

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TU Delft AM2520-P Philosophy of mathematics

januari 2024

Abstract

An informal treatment of meaning in mathematics is presented. Searle's Chinese Room experiment in the context of the Philosophy of Mind shows mathematics as manipulation of formal symbols is problematic for the mathematician. Understanding of mathematics cannot exist without meaning in mathematics. The problem of meaning in mathematics is introduced: how can we reconcile mathematics with meaning? To this end, we equate meaning in mathematics with the concept of intentionality. Then, we examine meaning in the context of major philosophies of mathematics: Platonism, formalism and intuitionism. Possible Platonist stances are criticised, and it is considered how Formalism and Intuitionism can accommodate for meaning as a part of mathematics. Lastly, we conclude that Intuitionism best incorporates meaning as part of mathematics, as it accounts for meaning in a "natural" way.

"Mathematics is not about numbers, equations, computations, or algorithms: it is about understanding."

— W.P. Thurston

1 Chinese room, French Scrabble

In his well-known 1980 paper[1], American Philosopher John Searle introduces his famous "Chinese Room Experiment" (CRE) to argue against what he calls "strong artificial intelligence" [2]. In this now famous thought experiment Searle imagines being locked in a room with three batches of Chinese symbols. Subsequently, he is handed a set of instructions on matching symbols to other symbols. Unable to read Chinese, Searle views the symbols as formal, that is, having no

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[1] Searle, J. R. (1980). Minds, brains, and programs. *Behavioral and brain sciences*, 3(3), 417-424.

[2] The notion that, given the right program, a computer can be said to understand things.

meaning to him whatsoever. Fortunately, the instructions are in English, a language he understands perfectly. Unbeknownst to him, the people who gave him the three batches and the set of instructions call the first batch "a script", the second batch "a story", and the third batch "questions". The instructions also further instruct Searle to hand back certain symbols in a certain order in reply to the questions, and these are dubbed "answers to the questions". The sets of rules are called "the program". If the story given to him would be in English, Searle can answer the questions without issue, it is his native language after all. It is not hard to imagine that Searle, after been in the room for a long time, becomes really good at answering questions in Chinese, and the program is also constantly improved. From the viewpoint of the outside observer, Searle is perfectly capable of answering questions about a story in Chinese, and thus he seems to understand Chinese. His answers to Chinese and English questions would seem equally good. However, in these two cases, something else entirely is going on. In the case of English, there is a supposed mind processing and understanding the questions and answering them. In the Chinese case, Searle's ability to answer questions is due to pure manipulation of formal symbols. Nowhere can we presume to claim that Searle really understands Chinese. However problematic, the room manages to pass the Turing test.

The CRE has elicited many responses[3], and critisisms[4] and has been a staple in the discourse around artificial intelligence. However, apart from the context of AI, it invites us to question if the room can explain human understanding[5]. In a sense, it is the exact inverse of the AI question. If we are to deny the person inside room the property of understanding things, how can we justify understanding anything ourselves? In this sense, the Chinese Room has its place in the philosophy of mind[6]. How does a human speaking Mandarin Chinese merit being prescribed "an understanding of Chinese" as opposed to (the person in) the room? Is understanding a language not being able to produce formal manipulation of symbols, thoughts or some other objects?

In 2015, Nigel Richards won the French language Scrabble Championship in Belgium[7]. This is certainly a remarkable feat for any French speaker. However, Richards is from New-Zealand and does not speak any French whatsoever. Richards managed to accomplish his victory solely by memorization of words

[3]see the end of Searle's paper for many short commentaries of contemporary thinkers.

[4]See for instance: Rey, G. (1986). What's Really Going on in Searle's "Chinese Room". *Philosophical Studies: An International Journal for Philosophy in the Analytic Tradition*, 50(2), 169-185.

[5]As Searle says: "Now the claims made by strong AI are that the programmed computer understands the stories and that the program in some sense explains human understanding. But we are now in a position to examine these claims in light of our thought experiment [CRE]." Searle (1980), p.418

[6]The branch of philosophy that explores the nature of the mind, consciousness and the relationship between the mind and the body.

[7]Chappell, B. (2015). Winner of French scrabble title does not speak French. npr. July 21. Available at
`https://www.npr.org/sections/thetwo-way/2015/07/21/424980378/winner-of-french-scrabble-title-does-\not-speak-french?t=1612259812093`, consulted 3/1/24.

from a French dictionary, reportedly by repeating words to himself while riding his bicycle. How does this relate to the CRE? One could say Richards himself operates during the tournament as some kind of internalised[8] Chinese Room, being able to manipulate the abstract patterns and symbols of French which have no meaning to him.

To the bystander at the Scrabble board, it would seem that Richards is a French speaker. However, if understanding is to mean anything at all, we should be mistaken to conclude that Richards in fact understands French simply from his ability to manipulate the formal symbols of the French alphabet. It seems that some kind of internalized meaning is missing. Not only must the French words be defined in a dictionary outside of Richards mind, for him to speak French, we need the words to be about something to Richards. Certainly, to humans involved in any kind of mental activity (e.g. mathematics), it would be a waste to attribute understanding to something as plain as internalisation of formal processes: a piece of the puzzle is missing.

Figure 1: Nigel Richards, Scrabble champion extraordinaire, Chinese Room in the flesh.

Searle's answer is clear: the difference lies in *meaning*. For the person in the room, the formal symbols are *meaningless*. For Richards, there is no more meaning in "échecs" than there is to "$\mathfrak{m}$"[9]. To a native speaker of Mandarin or French, the symbols have meaning. As Searle says in the same article: *"In the linguistic jargon, they* [the symbols] *have only a syntax but no semantics."* (p. 422). We conclude that we cannot reduce understanding down to manipulation of formal symbols: semantics is an indispensable part of comprehension. If we neglect internalised semantics, the person in the CRE can speak Mandarin and Richards is fluent in French, and the student of mathematics who simply memorizes meaningless formulas understands mathematics.

2 The problem of meaning in mathematics and intentionality

As hinted at above, the discussion about meaning should be of great concern to the mathematician. Presumably, the mathematician should hope to claim she understands (some parts of) mathematics. If she does not, what is the

[8] When asked about how Richards does it, fellow Scrabble player James Fatsis replied: "Basically, what he does is, he looks at word lists and looks at dictionary pages... he can conjure up the image of what he has seen. He told me that if he actually hears a word, it doesn't stick in his brain. But if he sees it once, that's enough for him to recall the image of it. I don't know if that's a photographic memory; I just think it's something that his brain chemistry allows him to do." (see 7)

[9] The TEX-code for this symbol is taken from `https://tex.stackexchange.com/questions/487943/a-strange-latex-symbol`

point of her toils? What would the whole of mathematics be working towards if not understanding of certain "objects"? As the previous section shows, even with internalised manipulation of formal symbols, without meaning there can be no understanding. This is further complicated by the prevalence of formal symbolism in mathematics, often seen as one of its fortes. To make things worse, mathematics' seemingly "objective" nature seems to leave no room for meaning as an inseparable part of mathematics: a correct proof is a correct proof, whether it is you or me who wrote it. Where is meaning, a seemingly "subjective"[10] notion in all of this? This is what we will call the problem of meaning in mathematics: how can we reconcile mathematics with meaning?

Meaning in mathematics is not trivial. In education, it is rarely mentioned, if at all. In the field of applied mathematics, we might find meaning in the way that mathematics is about helping us in the construction of a bridge, in predicting the market or solving a problem in cryptography, however, these are not what the field of mathematics is about. The vast majority of work in the field of mathematics is not applied. Research in mathematics is not always done with intention of helping humanity directly in some way. Applications of mathematics are not a an inherent part of mathematics. It exists without bridges, without markets and without cryptography. This might be the meaning of the field of mathematics itself. However, we are interested in the meaning within mathematics: the meaning of the subjects of its study. Meaning is certainly related to semantics. But this relation can only take us so far, the semantics[11] of logical symbols themselves might be included in mathematics themselves, which does not solve our problem of meaning. Whatever meaning is in mathematics, is highly disputed. Mathematicians and philosophers alike are not even sure on what mathematics is[12], let alone what it is about or what it means to do mathematics. We will not attempt to explain the purpose or meaning of mathematics. All that is certain that meaning in and of mathematics is mathematics being about something, and this "being about something" cannot be left out. The aboutness can be seen as precisely the role of meaning in mathematics. Aboutness can manifest itself in the form of perceived importance, seriousness, or descriptiveness of mathematics. Mathematics has something to say. This maybe exemplified when Hardy famously writes about chess and the "importance" of mathematics:

> *"A chess problem is genuine mathematics, but it is in some way 'trivial' mathematics. However ingenious and intricate, however original and surprising the moves, there is something essential lacking. Chess problems are unimportant. The best mathematics is serious as well as beautiful —important if you like, but the word is very ambiguous, and 'serious' expresses what I*

[10] Here, we take objective to mean observer-independent, subjective as observer-dependent.

[11] the discussion of meaning in the context of logic is certainly of interest. However, this is a more technical discussion of mathematics. For the sake of simplicity and the fact that the author is not familiar enough with this field, we will not go into the realm of logic.

[12] An interesting one page(!) opinion piece is Hersh, R. (1998). What is mathematics, really?. *Mitteilungen der Deutschen Mathematiker-Vereinigung*, 6(2), 13-14.

mean much better."[13]:

To attempt to answer more precisely what role meaning plays in mathematics, we shall look further into the notion of *intentionality*, which is at the heart of the Philosophy of Mind. Meaning itself might be a term that is too broad for our purposes, and in the philosophical setting we will prefer working with the notion of intentionality. Whereas "meaning" might have multiple definitions, intentionality admits a philosophical definition that is be easier to wield. In the following, we use the concepts interchangeably, knowing that behind the word "meaning" is the concept of intentionality. Intentionality is perhaps best explained in the following way by McIntyre and Smith:

> *"Many, perhaps most, of the events that make up our mental life - our perceptions, thoughts, beliefs, hopes, fears, and so on - have this characteristic feature of being 'of' or 'about' something and so giving us a sense of something in our world. When I see a tree, for example, my perception is a perception* ***of*** *a tree; when I think that* $3 + 2 = 5$*, I am thinking* ***of*** *or* ***about*** *certain numbers and a relation among them; when I hope that nuclear war will never take place, my hope is about a possible future state of the world; and so on. Each such mental state or experience is in this way a* ***representation*** *of something other than itself and so gives one a sense of something. This representational character of mind or consciousness - its being 'of' or 'about' something - is 'intentionality'."*[14]

In short, intentionality is the "aboutness" of our mental activities. Mathematics is about something. In doing mathematics we are manipulating formal objects but never without semantics. In this way, we can view meaning in mathematics as intentionality of these manipulations. At this point, it is not clear what role intentionality plays in mathematics. In exploring the subject, it will be helpful to do so through the lenses of the major schools in the philosophy of mathematics: Platonism[15], Formalism and Intuitionism[16].

3 Meaning in Platonism

Platonism is the school of philosophy of mathematics that seems have proven itself the most resilient over the years. It is characterized by the notion that mathematical objects exists and mathematicians discover them. In this sense, mathematics is like physics, where mathematicians discover laws and patterns governing these objects. We should not be mistaken that mathematical reality exists as physical objects. Rather, Platonism stresses that they exist outside of space and time. Good mathematics would then consist of good descriptions. In

[13] Hardy, G. H. (1992). A mathematician's apology. *Cambridge University Press.* p. 20

[14] McIntyre, R. T. and Smith, D. W. (1989). Theory of intentionality. *Center for Advanced Research in Phenomenology.* p.1

[15] Or Realism, if you like.

[16] Of course, this is but a sample of schools in philosophy of mathematics, other major schools include Nominalism, Logicism and Constructivism.

the view of Platonism, we regard the objects subject to mathematics as real. A good description is given by Hardy:

> *"For me, and I suppose for most mathematicians, there is another reality, which I will call 'mathematical reality'; and there is no sort of agreement about the nature of mathematical reality among either mathematicians or philosophers. Some hold (...)*[17] *that it is outside and independent of us. A man who could give a convincing account of mathematical reality would have solved very many of the most difficult problems of metaphysics. If he could include physical reality in his account, he would have solved them all. (...) I believe that mathematical reality lies outside us, that our function is to discover or observe it, and that the theorems which we prove, and which we describe grandiloquently*[18] *as our 'creations'; are simply our notes of our observations. This view has been held, in one form or another, by many philosophers of high reputation from Plato onwards (...)"*[19]:

Platonism is plagued by several problems, and has been critised extensively[20]. Hardy himself admits that we are missing a convincing account of mathematical reality. If mathematical objects are real but do not exist in space and time, where are they? If mathematical objects exist outside of us, how come we do mathematics inside our mind? This gap between the mental and the realm of objects can be seen as the major problem of Platonism. There is a sense in which Platonism is more like a religion[21] than a philosophy, claiming the existence of a realm inaccessible to the mathematician. On the wall of the cave we then see only the Platonic shadows of these perfect objects stored away in a supposed mathematical heaven. Admittedly, this assessment is rather unfair to the Platonist. Many great mathematicians[22] have been Platonist in one form or another. Platonism admits to great aesthetic and functional value: mathematics as a discovery of some higher truth or form of existence is the great motivating factor in mathematics for some. Due to the burden of proof of claimed existence of mathematical objects, Platonism in mathematics can be considered an easy target for the philosopher of mathematics. However, it should be noted that in actually doing mathematics we cannot avoid being Platonists. This amusing fact can be proven by sitting in any lecture on mathematics and hearing a delightful discussion of all kinds of forms such as differential equations, graphs (both kinds), or differentiability, just as you would hear about plants, trees, or DNA in any biology lecture.[23] Even the most anti-Platonist mathematician will find himself discussing the most mathematical of objects as if they were

[17] Hardy considers here the notion that the objects are mental also as Platonist, although for our purposes, this is not a part of Platonism, rather Constructivism. The part left out here reads: "... that it is 'mental' and that in some sense we construct it, others ..."

[18] Ironically, this means "pompous or extravagant in language".

[19] Hardy, G. H. (1992). A mathematician's apology. *Cambridge University Press.* pp. 63-64

[20] See for example: Balaguer, M. (2001). Platonism and anti-platonism in mathematics. *Oxford University Press*, USA.

[21] which is not a pejorative

[22] Leibniz, Cantor and to a certain extent Gödel might be considered Platonists

[23] The authors ignorance on the field of biology may or may not be painfully obvious here.

"real" in the sense of Hardy. Platonism is just incredibly practical to accept for any practitioner of the subject. This paradoxical nature of the anti-Platonist stance makes Platonism attractive for many mathematicians. This shows the importance of reconciling Platonism with meaning.

We shall consider meaning in Platonism in two ways[24]:

- Meaning can be incorporated into the realm of mathematical objects, this we shall call "inclusive" Platonism. It might seem peculiar to include a concept so broad into the Platonist realm of objects. However, we should consider that not only mathematical "nouns" live in this realm. It might be immediate that triangles and circles and π live in this realm, but so do theorems and corollaries. A mathematician refers to "The Riemann-Lebesgue Lemma" as if it were a really existing object, just as she might refer to an "isosceles triangle". Now, if theorems and concepts are part of this realm, why would we draw the line at meaning? Thus we can include meaning among the objects of mathematics.

- Meaning should be excluded from the realm of Platonist objects. Meaning or intentionality itself could operate on a meta-level, where intentionality is about the interaction of the mind and the Platonist realm. Intentionality is not a part of this realm, but it is a governing structure of this realm. We might then call such a view "exclusive" Platonism.

Several objections to inclusive Platonism might be raised immediately. At the very least, it should be clarified whether every mathematical object carries its own intentionality in a pair, or that intentionality as a whole is included. More generally, how does meaning correlate to the object? Intentionality itself seems to be about mathematical forms, but also about mathematics itself. This dual nature makes it difficult to imagine intentionality or any other interpretation of meaning as a part of the realm of ideas. Statements about mathematics being difficult, beautiful, complex are on a same "meta" level as meaning, but would we then also include these as Platonic objects? It is not clear that statements about mathematics[25] should be a part of mathematics. That would seem rather problematic, and we might end up in an infinite regression. If the description of mathematics is part of mathematics, the description of this description is again part of mathematics. This would make everything mathematics. This boils down to a "slippery slope" argument: for the inclusive Platonist, where do the inclusions stop? Mathematics is supposedly of perfect forms that exceed human existence, so then why include meaning and and beauty, such very human concepts?

Exclusive Platonism, although not having the immediate problem of infinite regression, still faces the same burden of proof as mentioned before. We might

[24] We should not claim that these are the only two ways in which Platonism might accomodate intentionality, and must be careful not to create a false dichotomy. Anyone serious about Platonism is welcome to explain their way to incorporate meaning.

[25] The "metalanguage" if you will.

factor meaning in mathematics into the mind-body problem[26]: meaning is the way it is interesting for us to interact with the forms. It is not immediate how meaning can be accommodated for by exclusive Platonism. Meaning as a cupola over the platonic realm, correlating certain objects might be the most generous interpretation.

The main problems of Platonism remain[27], whether we include meaning or not. However, its inevitability warrants a deeper analysis of meaning in Platonism. For now, the place of mathematical meaning remains a mystery in Platonism.

4 Meaning in Formalism

In the formalist[28] school, we view mathematics as a "game" of symbols and logic. Mathematics as an activity emerges from the manipulation of formal symbols and rules of inference. The formalist view is that the existence of mathematical object is entirely irrelevant for doing mathematics: metaphysics should be avoided, logical structures themselves will constitute mathematics. As philosopher Alan Weir puts it:

> *"One common understanding of formalism in the philosophy of mathematics takes it as holding that mathematics is not a body of propositions representing an abstract sector of reality but is much more akin to a game, bringing with it no more commitment to an ontology[29] of objects or properties than ludo[30] or chess."*[31]

When playing a game, say Chess, we are neither burdened nor concered with the existence of "pawnness", "knightness" or "queenness", we are simply playing a game, in which we are given a set of rules and a way to make legal moves. The pieces on the board are made out of matter, and we attach to them rules. Mathematics exists on the paper in the sense that Chess exists on the board. Now, we can do mathematics without paper, just as we might do Chess without the board. The concepts and rules are what count. In this view, it makes no more sense to ask a mathematician about the existence of differentiability than to ask Magnus Carlsen about the existence of the checkmate. Checkmate is a formal concept that only makes sense within the context of the game of chess

[26] The fundamental question of how mental states (such as thoughts, emotions, and perceptions) are related to physical states or processes in the body.

[27] Our arguments against inclusive and exclusive Platonism are admittedly strawman arguments, since these are not viable positions found in the literature. The argument here is simply to show that meaning is not a given part of Platonism per se.

[28] Often associated with Hilbert and his famous program.

[29] Meaning "existence".

[30] A strategy board game for two or four players.

[31] W, Alan, "Formalism in the Philosophy of Mathematics", *The Stanford Encyclopedia of Philosophy (Spring 2022 Edition)*, Edward N. Zalta (ed.), `https://plato.stanford.edu/archives/spr2022/entries/formalism-mathematics/` consulted 4/1/24. Weir calls this formalism "game formalism" (as opposed to many other types of formalism).

and is a relation between legal moves in the game. Differentiability is a formal concept that only makes sense in the context of mathematics.

Meaning as inseparable part of mathematics might pose a direct threat to formalism. Formalism does not allow for existence of external objects or notions, how then will it allow meaning? Meaning, as a subjective notion surely seems to go against the nature of formalism, which strives for strict logical views on the nature of mathematics. Meaning in the case of the language game view can be considered to be one of the emergent properties of such a game. Meaning maybe arises in mathematics as a game the same way a the meaning of a knight is determined by the way we use it.

The analogy between mathematics and chess in the context of formalism is further elaborated upon in the work of Landsman and Singh. They propose the view of mathematics as what they call "a motley of language games of a very specific (formal) kind"[32] These language games are games in the sense of Wittgensteinian family resemblances[33]. They define such a language game as having the following characteristics[34]:

1. A language game is a practice where certain words and symbols are used (operationally)

2. This use is determined by specific rules (forming the grammar of the language game)

3. The meaning of (most) words and symbols is in turn determined by their use.

Mathematics as a language game has additional properties besides these ones. Most notably, mathematics is open: new elements can be added to it. In this way, meaning is an inherent part of mathematics. Without it, we cannot play the game of mathematics. This is precisely what we want. Meaning is as much mathematics as any proposition, and in this sense, is even more fundamental. The knight in chess is defined solely by the way we use it. Without resembling an actual physical piece, we can use the knight in an abstract game of chess: in fact, a knight is the way we use it. The way we use and move the knight constitute its meaning. Much better than Platonism, we find meaning as an inseparable part of this formalist view of mathematics.

5 Meaning in Intuitionism

We will now consider the place of meaning in Intuitionism as popularised by the writings of Brouwer, influenced by Kant. Intuitionism might best be explained

[32]Landsman, K., & Singh, K. (2023). Is mathematics a game?. arXiv preprint arXiv:2311.12478. p.10.

[33]Meaning the way in which members of a category share overlapping features or characteristics, originating in the later philosophy of Wittgenstein

[34]Landsman and Singh, p.7

as being in opposition to Formalism. We can characterise Intuitionism as being constructivist in nature. Mathematical objects can only exist in the mind, where they are constructed by the mathematician. In no way do mathematical subjects of our thought exist in a separate dimension apart from ours. This immediately highlights the subjectivity of mathematics. Mathematical truths can only be reached by construction of explicit proofs and structures, where this construction is in the individual mathematicians mind. In this view, intuition and construction of mathematical ideas are central. In Brouwer's words:

"The question where mathematical exactness does exist, is answered differently by the two sides; the intuitionist says: in the human intellect, the formalist says: on paper.""[35]

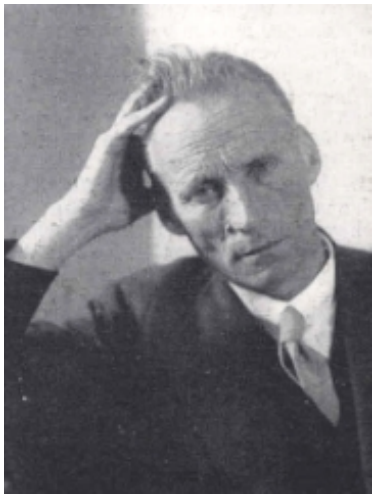

Figure 2: L.E.J. Brouwer, Dutch mathematician, father of modern mathematical Intuitionism

Meaning and mathematics are direct constructs of the intellect, and thus can presumably be taken together without too much objection. The proofs we construct, the definitions we wield; these constructions give inherent meaning to mathematics. Intuitionist concepts of meaning can thus arise directly from construction of mathematical concepts and proofs. If they were not meaningful, we would not construct them. We cannot construct mathematics that is not about something, since all perceptions of the mind are subject to intentionality. I think of mathematics, therefore it has meaning to me. The knowledge of mathematics can only grow by experiencing and constructing more mathematics of a similar kind. This introspectiveness of intuitionist mathematics was highlighted by Detlefsen:

"According to Brouwer, mathematics is essentially a form of introspective constructional activity or experience whose growth or development thus cannot proceed via the logical extrapolation of its content (as classical epistemology maintains), but rather only by its phenomenological or experiential development-that is to say, its extension into further experience of the same epistemic[36] *kind."*[37]

[35] Brouwer, L. E. (2000). Intuitionism and formalism. *BULLETIN-AMERICAN MATHEMATICAL SOCIETY, 37(1), 55-64.* p. 56

[36] Meaning anything related to knowledge, understanding, or the ways in which we acquire and justify beliefs.

[37] Detlefsen, M. (1990). Brouwerian intuitionism. *Mind*, 99(396), 501-534. p.515

One of the problems of Intuitionism might be that mathematics now becomes subjective. However, to neglect Intuitionism for this reason is to deny the social aspect of mathematics. Two intuitionist mathematicians can agree through argument and careful clarification of concepts. In the acquisition of mathematics, discussion and dialogue is not only beneficial, but unavoidable. This social aspect also encompasses reading mathematical literature. The book's author proposes and suggests mathematical constructs and presents them to the reader. The reader is invited to then construct his own, and see that his intuition agrees if the mathematics is to be correct. The symbols in the book are an invitation to construct. The internalisation of meaning of symbols arises through this reading, constructing, and weighing against one's own intuition.

Intuitionism seems to admit to a natural[38] and simple incorporation of meaning in mathematics. In the sense of Intuitionism, we can incorporate meaning in the sense that symbols and their manipulation mean the same to mathematicians, where this meaning comes from their intuition and careful discussion of their construction. This meaning arises in the same way meaning arises from symbols in a language: from experience-based inference, intuition, careful discussion and social consensus.

6 Conclusion

We have seen that the CRE and the case of Nigel Richards prompt us reevaluate what understanding can mean for the mathematician. We asses that one cannot infer understanding of mathematics from mere formal manipulations of symbols. Meaning seems to present itself as inseparable from mathematics itself. The problem of meaning in mathematics is precisely the question of how we reconcile mathematics with meaning. We characterised meaning (perhaps controversially and naively) as intentionality. We visited the schools of Platonism, Formalism and Intuitionism and explored how we might incorporate meaning in each. We conclude Platonism cannot solve the problem of meaning, or at the very least it remains to be explained how. Game formalism has meaning as one of its fundamental parts. And might be more accurate as a description of the field of mathematics in this regard. Intuitionism naturally accounts for how meaning can be a part of mathematics, where meaning arises through careful construction of mathematical objects. In view of Intuitionist thinking, we can regard the meaning of mathematical symbols as originating in the "social" aspect of mathematics. It must be formed through consensus of mathematicians.

Admittedly, we have considered meaning in a non-rigorous way, hence further investigation of meaning in the context of semantics in linguistics and logic is warranted. Here, a more technical foundation may be laid for meaning in mathematics. Let us then ponder carefully the meaning in, of and about our philosophical and mathematical endeavors! [39]

[38] Might we say intuitive?

[39] The image of Richards was taken from `https://en.wikipedia.org/wiki/Nigel_`

Richards_(Scrabble_player)#/media/File:Nigel_Richards_2018.jpg and is taken by user Kingoflettuce. The image of Brouwer was taken from https://resources.huygens.knaw.nl/bwn1880-2000/lemmata/bwn2/brouwerle